ASCII MASTER FLOW — PANEL 1/12: Glacier budget

```
ACCUMULATION -> snowfall, avalanche and refreezing
ELA -> annual gain equals annual loss
ABLATION -> melt, sublimation and calving where relevant
NET BALANCE -> geometry responds with a lag
MUST REMEMBER: Glaciation proceeds through accumulation-ablation balance, ice
  deformation/basal motion, erosion, transport and deposition; cirque–arête–horn, U-shaped
  valley, moraine and outwash logic must connect to Himalayan mass balance, proglacial lakes
  and GLOF cascades.
```

ASCII MASTER FLOW — PANEL 2/12: Retreat paradox

ICE PARTICLES -> continue moving downslope
TERMINUS -> receives forward ice flux
ABLATION > DELIVERY -> snout shifts up-valley
METRIC RULE -> flow direction differs from margin change

ASCII MASTER FLOW — PANEL 3/12: Erosion suite

```
PLUCKING -> jointed blocks removed
ABRASION -> polish, striae and rock flour
CIRQUE / ARETE / HORN -> headwall system
U-TROUGH / HANGING VALLEY -> valley-scale overdeepening
```

ASCII MASTER FLOW — PANEL 4/12: Fjord-ria firewall

FJORD -> glacial trough later flooded by sea
RIA -> river valley later drowned by sea
FJORD CUES -> deep water, steep walls, hanging valleys
RULE -> inundation preserves different pre-existing valleys

ASCII MASTER FLOW — PANEL 5/12: Depositional sorting

```
TILL / MORAINE -> direct ice, generally unsorted
ESKER -> meltwater ridge in or under ice
OUTWASH -> braided meltwater, relatively sorted
DRUMLIN -> streamlined drift/till landform
```

ASCII MASTER FLOW — PANEL 6/12: Ice-mass family

GROUNDED -> glacier, ice cap, ice sheet
FLOATING -> shelf, iceberg and sea ice
SEA LEVEL -> grounded loss adds ocean mass
BUTTRESSING -> shelf loss can accelerate inland discharge

ASCII MASTER FLOW — PANEL 7/12: Himalayan map hooks

GANGOTRI -> Bhagirathi
ZEMU -> Teesta
BARA SHIGRI -> Chenab system
SIACHEN -> Nubra-Shyok-Indus system

ASCII MASTER FLOW — PANEL 8/12: Water-timing sequence

EARLY LOSS -> local melt contribution may rise
STORED ICE DECLINES -> buffering weakens
LEAN SEASON -> reliability risk grows locally
QUALIFIER -> monsoon, snow and groundwater vary by basin

ASCII MASTER FLOW — PANEL 9/12: Glacial-lake test

```
BASIN / DAM -> how water is impounded
SLOPE + ICE -> possible displacement trigger
DAM CONDITION -> overtopping, piping or erosion
DOWNSTREAM -> exposure converts hazard into risk
```

ASCII MASTER FLOW — PANEL 10/12: GLOF cascade

```
TRIGGER -> slope failure, inflow or instability
WAVE / OVERTOPPING -> dam erosion begins
BREACH -> sudden release + sediment
VALLEY -> constrictions and assets shape consequences
CLOSE DISTINCTION: Snowline, equilibrium-line altitude, firn line and glacier terminus are
  distinct; till is unsorted ice-laid sediment while outwash is meltwater-sorted, and a fjord
  is a drowned glacial trough rather than any steep inlet.
```

ASCII MASTER FLOW — PANEL 11/12: Risk-reduction stack

```
SATELLITE INVENTORY -> changing lakes and slopes
FIELD ASSESSMENT -> dam material and thresholds
WARNING + DRILLS -> last-mile action
LAND USE + DESIGN -> reduce exposure and fragility
EVIDENCE LIMIT: Retreat varies by glacier and period; lake growth alone does not prove
  imminent outburst, while dam material, freeboard, slope instability, ice/rock avalanches and
  downstream exposure condition risk.
```

ASCII MASTER FLOW — PANEL 12/12: Cryosphere answer spine

```
DEFINE -> glacier, lake, GLOF and risk
MAP -> glacier-basin relation
EXPLAIN -> budget, trigger and breach
QUALIFY -> heterogeneity, attribution and data date
```